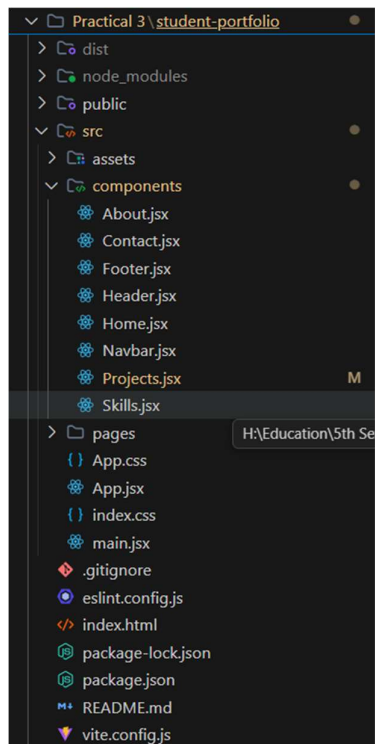


Practical 3 : API Integration and Data Rendering in React

Problem Definition :

- Integrate a public REST API (e.g., GitHub API) into the portfolio application to fetch and display a list of repositories dynamically.
- Use the Projects page built in Practical 2 as the integration point.
- Implement a loading spinner component shown while the request is in progress.
- Implement an error message component shown if the API call fails.
- Render at least the repository name and URL for each item returned.

Folder Structure:



Code:**About.jsx:**

```
function About() {  
  
  return (  
  
    <section className="section">  
  
      <h2>About Me</h2>  
  
      <p>  
        my name is rudra and i am a student.  
      </p>  
  
      <div className="details">  
  
        <p><strong>College :</strong> DEPSTAR </p>  
  
        <p><strong>Subject :</strong> ITUE301 - ADVANCED WEB DEVELOPMENT  
FRAMEWORKS</p>  
  
        <p><strong>Semester :</strong>5th Sem</p>  
  
        <p><strong>Enrollment No :</strong> D25DIT081</p>  
  
        <p><strong>Division :</strong> 5IT </p>  
  
      </div>  
  
    </section>  
  
  );  
}  
  
export default About;
```

Contact.jsx:

```
import { useState } from "react";  
  
function Contact() {  
  const [message, setMessage] = useState("");  
  
  return (  
    <section className="section">  
      <h2>Contact Page</h2>  
      <p>Send a message below and see real-time updates:</p>
```

```

    <div className="details" style={{ marginTop: "25px" }}>
      <label htmlFor="messageInput" style={{ display: "block", marginBottom: "10px",
color: "#bdbdbd" }}>
        Write Message:
      </label>
      <input
        id="messageInput"
        type="text"
        value={message}
        onChange={(e) => setMessage(e.target.value)}
        placeholder="Type your message"
        style={{
          padding: "12px 20px",
          width: "100%",
          maxWidth: "450px",
          borderRadius: "8px",
          border: "1px solid #333",
          background: "#1b1b1b",
          color: "#fff",
          fontSize: "16px",
          outline: "none"
        }}
      />

    <div style={{ marginTop: "25px" }}>
      <p><strong>Entered Text (Real Time):</strong></p>
      <p style={{ color: "#fff", marginTop: "10px" }}>
        {message || "No text entered yet..."}
      </p>

      <div style={{ marginTop: "15px" }}>
        <p><strong>Character Count:</strong> {message.length}</p>
      </div>
    </div>

  </div>
</section>
);
}

export default Contact;

```

Footer.jsx:

```
function Footer(props) {  
  
  return (  
  
    <section className="footer">  
  
      <h2>Contact</h2>  
  
      <p>Email : {props.email}</p>  
  
      <p>Phone : {props.phone}</p>  
  
      <p>City : {props.city}</p>  
  
      <hr />  
  
      <p>© 2026 Student Portfolio</p>  
  
    </section>  
  
  );  
}  
  
export default Footer;
```

Header.jsx:

```
function Header(props) {  
  
  return (  
  
    <section className="hero">  
  
      <h1 style={{ color: props.themeColor }}>{props.title}</h1>  
  
      <h2>{props.name}</h2>  
  
      <p>{props.role}</p>  
  
      <button>View Skills</button>  
  
    </section>  
  
  );  
}  
  
export default Header;
```

Home.jsx:

```
import { useState } from "react";
import Header from "./Header";
import About from "./About";
import Skills from "./Skills";

function Home() {
  const [showAbout, setShowAbout] = useState(true);

  const skills = [
    "HTML",
    "CSS",
    "JavaScript",
    "React",
    "Node js",
    "Vite"
  ];

  return (
    <div>
      <Header
        title="Student Portfolio"
        name="Rudra Maiyariya"
        role="Computer Engineering Student"
        themeColor="#ffffff"
      />

      <div style={{ textAlign: "center", margin: "10px 0" }}>
        <button onClick={() => setShowAbout(!showAbout)}>
          {showAbout ? "Hide About Section" : "Show About Section"}
        </button>
      </div>

      {showAbout && <About />}

      <Skills skillList={skills} />
    </div>
  );
}

export default Home;
```

Navbar.jsx:

```
import { NavLink } from "react-router-dom";

function Navbar() {
```

```

return (
  <nav className="navbar">

    <h2 className="logo">Portfolio</h2>

    <ul>

      <li>
        <NavLink to="/" className={{ { isActive }} => (isActive ? "active" : "")}>
          Home
        </NavLink>
      </li>

      <li>
        <NavLink to="/projects" className={{ { isActive }} => (isActive ? "active" : "")}>
          Projects
        </NavLink>
      </li>

      <li>
        <NavLink to="/contact" className={{ { isActive }} => (isActive ? "active" : "")}>
          Contact
        </NavLink>
      </li>

    </ul>

  </nav>
);
}

export default Navbar;

```

Project.jsx:

```

import React, { useState, useEffect } from 'react';

const Spinner = () => (
  <div style={{ textAlign: 'center', padding: '20px' }}>
    <div style={{
      display: 'inline-block',
      width: '40px',
      height: '40px',
      border: '4px solid rgba(0,0,0,0.1)',
      borderRadius: '50%',
      borderTopColor: '#3498db',
      animation: 'spin 1s ease-in-out infinite'
    }}></div>
    <style>
      {
        @keyframes spin {

```

```

        to { transform: rotate(360deg); }
      }
    `}
  </style>
  <p>Loading repositories...</p>
</div>
);

const ErrorMessage = ({ message }) => (
  <div style={{ color: 'red', textAlign: 'center', padding: '20px', backgroundColor: '#ffeaea',
borderRadius: '5px' }}>
    <h3>Failed to load repositories</h3>
    <p>{message}</p>
  </div>
);

function Projects() {
  const [repos, setRepos] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState(null);

  useEffect(() => {
    fetch('https://api.github.com/users/RudraMaiyariya/repos?sort=updated&per_page=3')
      .then((res) => {
        if (!res.ok) {
          throw new Error(`Error: ${res.status} ${res.statusText}`);
        }
        return res.json();
      })
      .then((data) => setRepos(data))
      .catch((err) => setError(err.message))
      .finally(() => setLoading(false));
  }, []);

  return (
    <section className="section">
      <h2>My Projects</h2>

      {loading && <Spinner />}
      {error && <ErrorMessage message={error} />}
      {!loading && !error && (
        <div className="details" style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fill,
minmax(300px, 1fr))', gap: '20px' }}>
          {repos.map((repo) => (
            <div key={repo.id} style={{
              padding: "20px",
              border: "1px solid #ddd",
              borderRadius: "8px",
              boxShadow: "0 2px 4px rgba(0,0,0,0.1)",
              backgroundColor: "#fff"
            }}>

```

```

        <h3 style={{ fontSize: "20px", marginBottom: "10px" }}>
          <a href={repo.html_url} target="_blank" rel="noopener noreferrer" style={{
textDecoration: 'none', color: '#0366d6' }}>
            {repo.name}
          </a>
        </h3>
        <p style={{ color: '#586069', fontSize: '14px', marginBottom: '15px' }}>
          {repo.description || 'No description available.'}
        </p>
        /*
        <div style={{ display: 'flex', gap: '15px', fontSize: '12px', color: '#586069' }}>
          {repo.language} && <span>🟡 {repo.language}</span>
          <span>🌟 {repo.stargazers_count}</span>
        </div>
        */
      </div>
    )))
  </div>
})
</section>
);
}

```

export default Projects;

Skill.jsx:

```

function Skills(props) {
  return (
    <section className="section">
      <h2>Technical Skills</h2>
      <div className="skills">
        {props.skillList.map((skill) => (
          <span className="skill" key={skill}>
            {skill}
          </span>
        ))}
      </div>
    </section>
  )
}

```

```
);  
  
}  
  
export default Skills;
```

NotFound.jsx:

```
function NotFound() {  
  return (  
    <section className="section">  
      <h2>404 - Page Not Found</h2>  
      <p>The page you are looking for does not exist.</p>  
    </section>  
  );  
}  
  
export default NotFound;
```

App.jsx:

```
import { BrowserRouter, Routes, Route } from "react-router-dom";  
import "./App.css";  
  
import Navbar from "./components/Navbar";  
import Footer from "./components/Footer";  
import Home from "./components/Home";  
import Projects from "./components/Projects";  
import Contact from "./components/Contact";  
import NotFound from "./pages/NotFound";  
  
function App() {  
  return (  
    <BrowserRouter>  
      <div>  
        <Navbar />  
  
        <Routes>  
          <Route path="/" element={<Home />} />  
          <Route path="/projects" element={<Projects />} />  
          <Route path="/contact" element={<Contact />} />  
        </Routes>  
  
        <Footer  
          email="rudra@email.com"  
          phone="+91 1234567891"  
          city="Porbandar"  
        />  
      </div>  
    </BrowserRouter>  
  );  
}
```

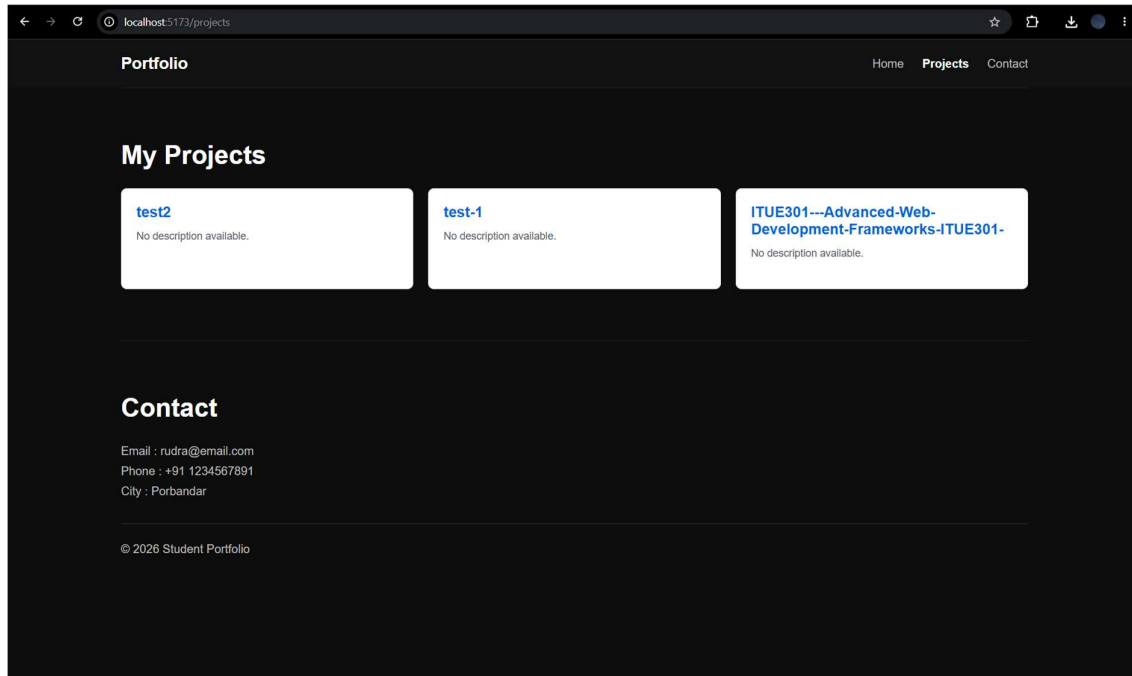
```
    </BrowserRouter>  
  );  
}
```

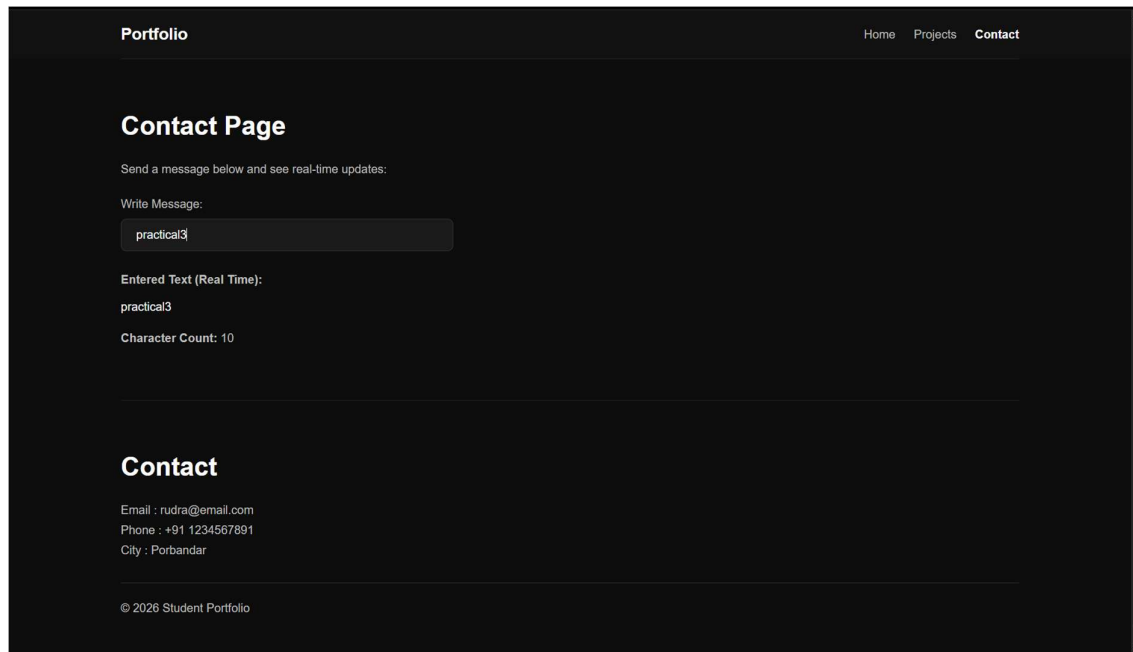
```
export default App;
```

Main.jsx:

```
import { StrictMode } from 'react'  
import { createRoot } from 'react-dom/client'  
import './index.css'  
import App from './App.jsx'  
  
createRoot(document.getElementById('root')).render(  
  <StrictMode>  
    <App />  
  </StrictMode>,  
)
```

Output:





Supplementary Problems :

- Add a retry button that re-triggers the fetch when an error occurs.
- Add a search input that filters the rendered repository list by name.
- Display the repository's star count alongside its name.

```
import React, { useState, useEffect } from 'react';

function Projects() {
  const [repos, setRepos] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState(null);

  // Supplementary: Search and Retry state
  const [searchTerm, setSearchTerm] = useState("");
  const [retryCount, setRetryCount] = useState(0);
```

```

useEffect(() => {
  setLoading(true);
  setError(null);

  fetch('https://api.github.com/users/RudraMaiyariya/repos')
    .then((res) => {
      if (!res.ok) throw new Error(`Error: ${res.status}`);
      return res.json();
    })
    .then((data) => setRepos(data))
    .catch((err) => setError(err.message))
    .finally(() => setLoading(false));

  // Adding retryCount to the dependency array makes it re-fetch when the retry button is
  clicked
  }, [retryCount]);

  // Supplementary: Filter the list based on the search input
  const filteredRepos = repos.filter(repo =>
    repo.name.toLowerCase().includes(searchTerm.toLowerCase())
  );

  return (
    <section className="section">
      <h2>My Projects</h2>

      {/* Supplementary: Search Input */}
      <input
        type="text"
        placeholder="Search repositories..."
        value={searchTerm}
        onChange={(e) => setSearchTerm(e.target.value)}
        style={{ padding: '8px', marginBottom: '20px', width: '300px' }}
      />

      {loading && <Spinner />}

      {error && (
        <div>
          <ErrorMessage message={error} />
          {/* Supplementary: Retry Button */}
          <button
            onClick={() => setRetryCount(prev => prev + 1)}
            style={{ padding: '10px', marginTop: '10px', cursor: 'pointer' }}
          >
            Try Again
          </button>
        </div>
      )}
    </section>
  );

```

```
    {!loading && !error && (  
      <div className="details" style={{ display: 'grid', gridTemplateColumns:  
'repeat(auto-fill, minmax(300px, 1fr))', gap: '20px' }}>  
        {filteredRepos.map((repo) => (  
          <div key={repo.id} style={{ padding: "20px", border: "1px solid #ddd",  
borderRadius: "8px" }}>  
            <h3><a href={repo.html_url}>{repo.name}</a></h3>  
  
            { /* Supplementary: Display star count */ }  
            <p style={{ marginTop: '10px' }}> ⭐ Stars: {repo.stargazers_count}</p>  
          </div>  
        ))}  
      </div>  
    )}  
  </section>  
);  
}
```

```
export default Projects;
```